

**Japanese vs. American Leadership Style of Youth Baseball Coaching:
A Quantitative Analysis of Player Performance Outcomes at the Professional Level**

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Abstract:

Youth baseball coaching in Japan adheres to a strictly autocratic leadership style of coaching that is intense and physically demanding, often involving discipline and punishment. American youth baseball coaching tends to follow a more participative and democratic leadership style, with coaches showing more care and encouragement for their players. To determine if coaching styles at the youth level have a measurable impact on performance outcomes at the professional level, a quantitative analysis of available baseball statistics is conducted, comparing the pitching, hitting and defensive metrics of professional U.S.-born players in Major League Baseball (MLB) and Japan-born players in Nippon Professional Baseball (NPB), with the ultimate goal of improving our understanding of effective coaching leadership styles. Findings do not support the hypothesis that the more disciplined style of Japanese youth baseball coaching has a measurable positive impact on performance outcomes of Japanese players at the professional level. Instead, the analyzed data indicates overall better performance metrics for the American players.

Introduction

Baseball has long been known as an American sport; after all, it is “America’s pastime.” But in recent years, the United States’ primacy in baseball has been challenged, especially with the influx of talent from Latin American countries such as Cuba, Venezuela, and the Dominican Republic. That being said, these countries do not challenge the status of Major League Baseball (MLB) as the premier league in baseball. However, one league does rival the MLB in popularity and player ability: Japan’s Nippon Professional Baseball (NPB) league. Despite the youth of Japanese baseball, Team Japan has placed much better than Team USA in the World Baseball Classic, the largest international baseball tournament. A potential explanation for Japan’s quick rise to baseball success is the style of coaching used at the youth level.

One difference between American and Japanese baseball cultures is the significantly more intense training regiments that Japanese players go through at the youth (or “little league”) level. In America, Little League baseball is a recreational activity enjoyed in the evenings and on the weekends. Coaches are volunteers, often the dads of players and others without professional coaching or baseball experience. As American Little League players age and improve, especially among those who show promise, the level of coaching improves. In Japan, there’s a different philosophy. The best coaches remain at the youngest level of play. Beyond that, American and Japanese youth baseball coaches exhibit different leadership styles. Americans tend to follow a participative leadership style of coaching (as defined by Castillo et al., 2014), with coaches showing more care, compassion, support, and encouragement for their players, and/or a democratic leadership style (as defined by Jin et al., 2022), in which the coach allows players to participate in the decision-making process. Japanese little league coaches practice an autocratic leadership style (as defined in the two studies cited above), in which the coach assumes all

authority, decision-making is centralized, and, especially in Japan, where the coach is domineering and the training is intense, mentally and physically demanding, often involving discipline and even corporal punishment (Sakuma, 2013). There is also a higher level of commitment demanded of Japanese youth players, with some programs requiring 12-hour practices every Saturday and Sunday, even on holidays (Milo, 2024).

If leadership “implies the existence of an organization in which a population responds to the leader’s guidance” (Ahlquist and Levi, 2011), then it could be instructive to see if and how well the youth baseball population in Japan responded to the guidance of its coaches when those players reached the professional level. This study will analyze the performance of tier 1 professional players in the United States and Japan to determine if the Japanese coaching style at the youth level gives Japanese players a distinct advantage over American players. Using youth baseball coaching as a model, information gained from quantitatively evaluating the effectiveness of one coaching style over another could be applied not just in sports but to the teaching of any skill.

Literature Review

Peer-reviewed studies on leadership coaching styles in sports other than baseball provide theoretical frameworks for the styles employed and, in some cases, provide empirical evidence for the effectiveness or ineffectiveness of autocratic vs. participative/democratic styles. For example, Castillo et al. (2014) shows that an autocratic coaching style has a significant effect in influencing the performance of young dancers who are still beginning to master a skill as autocratic coaching can affect the intrinsic motivation of an individual. To cite another example, Alferman et al. (2012) compared competitive youth swimmers from Germany and Japan on key

aspects of their psychological make-up, specifically their goal orientation, fear of evaluation, and perceived coach support. The Japanese swimmers perceived their training climates to be more competition-oriented than did the German swimmers, whereas the Germans reported significantly more instruction, positive feedback, and social support than the Japanese. The researchers concluded that “cross-cultural differences in perceived coach behavior may” account for the differences (Alferman et al., 2012). For example, failure in Japanese sports is often associated with a sense of shame, more so than in Western cultures (Heine et al., 2001). These findings strongly suggest that Japanese culture shapes and guides a more autocratic leadership coaching philosophy, and that Japanese youth baseball players are more tolerant of a coaching style that would be perceived as too demanding in America. As described by Miller (2011), baseball in Japan is aligned with Japanese sensibilities and Confucian values of discipline, selfless devotion, and sensitivity to hierarchy. This has been termed “Bushidō Baseball,” referencing *bushidō*, a samurai moral code concerning attitudes, behavior and lifestyle. However, since this is a strictly quantitative analysis and not a mixed-methods study, in which “richly descriptive culture-specific interpretations [may] account for cultural influences on leadership” (Stentz et al., 2012), the potential influence of *bushidō* will be noted here but not examined.

Methods

There are three ways to evaluate the difference of skill between Japanese and American baseball players: (1) Compare statistics between American and Japanese players in their respective native leagues. (2) Compare the performances (teams and players) of Team USA and Team Japan in the World Baseball Classic (WBC). (3) Examine the performances of foreign-born players commonly referred to as import players, i.e., professional Japanese baseball players who joined the MLB or professional American baseball players who joined the NPB. Within

each method, baseball can be broken down into three macro-level skill groups: offense, defense and pitching. When data permits, the data will be split into two categories: ability and discipline. Ability refers to a player's physical ability to perform relevant skills, such as hitting for power (doubles, triples, home runs), "stuff" (defined below), and outs above average. Discipline refers to a player's consistency and lack of mistakes. Relevant skills are errors, mistake pitches and on-base percentage. The distinction is made because the U.S. has a larger population than Japan and a much longer history in baseball. This means America has a larger pool from which to pull players, resulting in biological advantages that a smaller country like Japan cannot replicate. As reported in Kuhn (2015), the coach of Team Japan in the 2015 Little League world series stated that the American players were much larger than his team, meaning they will hit with more power and be able to throw harder.

The first method of evaluation is league statistics between American-born players in the MLB and NPB players. The first of the macro-level skill groups is pitching. The simplest metric for determining the quality of pitching is ERA¹, but ERA is highly dependent on who you are playing against. Therefore, MLB and NPB ERAs are not comparable as these pitchers are not facing the same batters. The same problem is also faced when looking at other top-performance tracking stat lines such as WHiP⁵ or K/BB⁶ rate, as they all track a pitcher's ability to best hitters. While every stat line is going to be subject to some form of bias, the least biased stats for pitching are those that measure a pitcher's physical abilities, not performance. Which will be broken down into two skill groups, "stuff" and control. "Stuff" is a conceptual category that refers to how hard it is to hit the pitcher's pitches. "Stuff" itself cannot be quantified but metrics such as velocity, spin rate and pitch movement can. "Stuff" metrics quantify ability. Control refers to the pitcher's ability to consistently place the ball where they want it to go. While control

can be quantified, that data is not available. But there are other metrics that can be used to track a pitcher's control: HBP⁷, WP⁸, and BK⁹. Control metrics quantify discipline. Velocity, spin rate and movement are the best pieces of evidence for determining the quality of a pitcher's "stuff." The more a ball spins, the quicker it crosses the plate, and the more it moves (X, Y axis), the harder it will be for the batter to hit it. However, due to limitations of available data, only velocity will be examined. For velocity, league-wide data was not available, but Crotin et al. (2021) tracked the pitching metrics of the pitching staffs from the MLB's Los Angeles Dodgers and the NPB's Hiroshima Toyo Carp from 2015-2019. This study included the tracking of fastball velocity.

The second macro-level skill group is defense. The best defensive stats for non-catchers are OAA¹⁰ and fld%¹¹. OAA is a StatCast metric so it cannot be used. Fld% can be used and tracks discipline since it tracks error rate. Double plays¹² are used in this analysis as they are effective statistics for tracking ability since they measure the ability to get the extra out. Another defensive metric used in this analysis are passed balls (PBs). PBs are the best non-StatCast metric for measuring a catcher's blocking (one of the five main catcher skills). Data for the other four skills was not available. Fld% and PBs track discipline, whilst OAA and DPs track ability.

The third macro-level skill group is hitting. Hitting suffers from the same fate as pitching in that most of the statistics are dependent on the skills of other players, namely pitchers and fielders. Unlike pitching, data that ignores this bias could not be found.

The second evaluation method is comparing the performance of Team USA and Team Japan in the World Baseball Classic (WBC). However, this comes at a very small sample size, as only five WBCs have been played. Furthermore, Team USA and Team Japan have only met four

times. Table 3 shows the performance record of these meetings. This is due to a cultural difference between the U.S. and other countries. The general consensus among American baseball players is that winning the World Series is much more important than winning the WBC, resulting in big stars not choosing the competition or not providing as much effort into winning. Many MLB teams bar players, usually pitchers, from playing in the WBC due to fear of injury. Moreover, the pitchers that are allowed to play in the WBC are given strict pitch counts and rules for how they can be played, which forces the Team USA manager's hand in managing the pitching staff. Therefore, WBC data will not be considered for this analysis.

The final method of evaluation is comparing the stats of import players. Data for players going from Japan to the United State is available; however, data for U.S. players going to Japan is not. Since all the players are in the same league, performance-based statistics can be used as well as StatCast data. For pitching, HBP, BK, and WP remain as the discipline variables, whereas the performance metric of ERA, WHiP, cFIB¹³, W/L, K/BB, and Inn/GS will serve as a hybrid of both ability and discipline. Japanese pitchers are expected to perform higher than the American players because only the best pitchers from Japan will be imported. For defensive data, fld% and a range for StatCast metrics will be used. For offense, basic hitting stats of BA¹⁴, OBP¹⁵, SLG¹⁶, and OPS¹⁷ will be used. Import players will also have their debut MLB year compared to their NPB statistics the season before. Pitching will use ERA and WHiP whilst hitting will use BA, OBP, SLG, and OPS.

Data Collection

For the purposes of this study, only the period of 2015-2022 will be examined. The year 2022 was chosen as the cut-off year because MLB introduced significant rule changes that year

that would confound data collection. Although MLB and NPB have always had different rulebooks, the 2023 MLB changes — the addition of the pitch clock, limited pitcher disengagement, and shortened base distances — altered game dynamics to favor hitters more than pitchers such that the American game is no longer as directly comparable with Japan's. Therefore, the exclusion of post-2022 data allows for a more equitable comparison of data as the rule differences between the leagues are smaller. The year 2015 was chosen as the start date because that was when MLB introduced StatCast, a new data-collection technology. StatCast utilizes a network of cameras and sensors in every MLB ballpark, allowing for significantly more data and advanced statistics to be collected. Importantly, StatCast only applies to the MLB but allows for a significant increase in available data. Despite being entirely quantitative, the data will be mixed independently. The data is collected from Stathead and Baseball Reference by Sports Reference, Baseball Savant, ProEyeKyuu, and The Lahman Baseball Database.

Findings

The pitcher control (discipline) metrics in Table 1 and Figure 1 found that there were ~1.77 times more hit batters per 9 innings completed by NPB pitchers than by U.S.-born MLB pitchers. NPB pitchers received ~1.54 times the number of BKs per 9 innings completed. And NPB pitchers threw ~1.48 times more wild pitches than U.S.-born MLB pitchers. Figure 2 shows that across the board between the minor and major leagues of the compared organizations, the Dodgers (MLB) had the higher fastball velocity. The defensive metrics in Table 2 find that NPB players turn fewer double plays, with ~0.96 times as many double plays per inning than in MLB. NPB catchers allow more passed balls, with ~2.21 times as many passed balls per inning than U.S.-born MLB catchers. NPB players have a higher fld% of ~0.08%. Looking at the control (discipline) metrics in Table 4, the Japanese pitchers hit fewer batters, balked less and had a

better SO/BB rate, but they threw more wild pitches. For performance-based metrics (hybrid of ability and discipline), the Japanese pitchers have a better ERA, W/L, cFIP, and last more innings, but a higher (and therefore worse) WHiP. As seen in Table 5, the Japanese import players have a lower fld%, below average OAA, arm strength, and overall fielding value¹⁸, but an above-average arm value¹⁹. Table 6 and Figure 3 show that Japanese import players get a hit ~0.26% of the time more than U.S. players and get on base ~8.22% of the time more than U.S. players. Japanese import players also have a higher OPS of ~0.078. American players have a higher SLG, but only by ~0.0042. Table 7 shows that of the Japanese import pitchers who played between 2015-2022 and played in the NPB the year before making their MLB debut, the average ERA increased by 1.61 and average WHiP increased by 1.63. Table 8 shows the same for hitters. Table 8 also shows that, with the exceptions of Shohei Ohtani's 0.24 increase in slugging and Nori Aoki's identical slugging, that statistics dropped across the board for Japanese import hitters.

Discussion

Import players are expected to perform better than the average U.S. player. After all, only the very best players from Japan play in the MLB. It is expensive to import a player (import tax, extra commodities such as translator), so only stars (e.g., Shohei Otani) are worth the investment. Therefore, import players should perform above the U.S. standard and it is only noteworthy when they do not.

The data analyzed herein does not support the hypothesis that the more-disciplined style of Japanese youth baseball coaching has a measurable positive impact on the performance outcomes of Japanese players at the professional level. For the most part, the analyzed data

indicates greater ability for American players. American fastball velocity was higher. Japanese import hitters experienced statistical declines after moving to the MLB and hit for less power (lower SLG) than their American peers. For pitchers, import pitchers performed worse when facing MLB hitters vs NPB hitters. For defenders, MLB players turned more double plays than NPB players, import fielders had below MLB average OAA, arm strength and fielding value.

In terms of discipline, NPB pitchers hit more batters, threw more wild pitches, and balked more than their U.S.-born MLB peers. In addition, import pitchers gave up more hits and threw more wild pitches than their U.S.-born counterparts. For defenders, NPB catchers allowed more passed balls. NPB defenders did have a higher fld% than their U.S.-born MLB counterparts, but by less than a 10th of a percent. Even so, the import players allowed more errors, with a fld% that is $\sim 2\%$ lower. Import defenders did hold an over-average arm value, but only by ~ 2 percentiles, which is insignificant based upon the sample size of six qualified defenders. For hitters, there is no data that makes a compelling argument either way.

Limitations

The data collected and analyzed here does not tell a full story. Other impactful data exists but was unable to be collected. The variables for control are also imperfect. HBP and WP do not tell the whole story of control. They just track when a pitcher misses by a lot. The batter's ability to get out of the way and the catcher's ability to block are also significant factors that disrupt the data. Balk data is biased by the difference of umpires between MLB and NPB. Fastball velocity metrics succumb to bias as NPB and MLB use different baseballs. The NPB ball is smaller with a better grip, which makes it easier for NPB pitchers to control the ball. The NPB strike zone is smaller than the MLB strike zone, which allows for a greater margin of error for pitch placement,

allowing pitchers to focus more on movement. Pitcher usage (as in how often and for how long a pitcher is expected to pitch) will also influence “stuff.”

Fld% is biased by the differences of ballpark size and batter tendencies. There is no way to determine how these biases affect play, but it is a possibility that, on average, a ball put in play will be easier or harder to field than that of the other league. Quality of shift is another potential bias. If a league on average is able to shift more effectively, then it will inflate the Fld%. Determining the quality of a shift is impossible without StatCast data. DPs are not without bias because the data is unable to consider the conversion rate in DP opportunities, i.e., unable to find data on DP conversion rate when a ground ball is hit to the infield when a DP is in order. DPs are usually assisted so data for MLB could not be sorted to just include U.S.-born players. A bias in passed balls is the unavailability of data on the quality (“wildness”) of the pitches thrown. In the cases of both PBs and DPs, innings played has been used as a substitute for both pitches and opportunities.

Conclusion

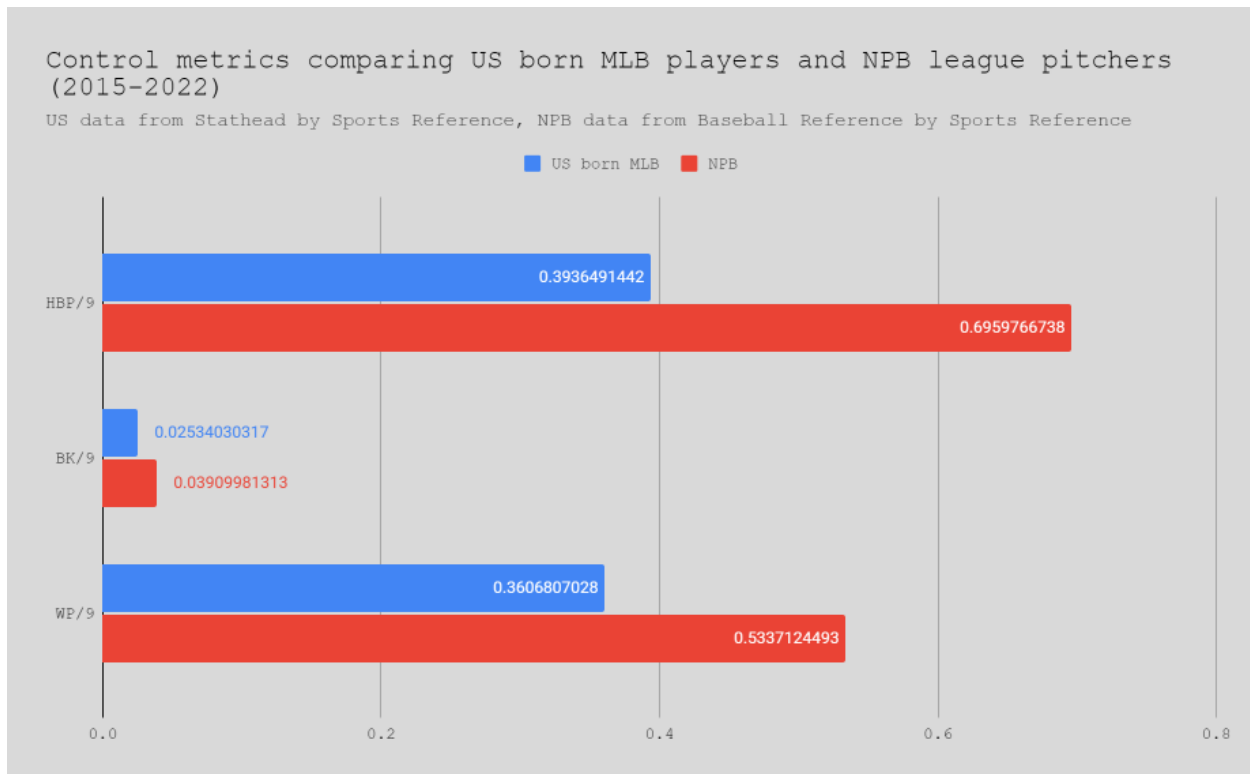
With the above caveats, the data analyzed does not support the hypothesis that the more disciplined style of Japanese youth baseball coaching has a measurable positive impact on the performance outcomes of Japanese players at the professional level. For the most part, the analyzed data indicates overall better performance metrics for American players.

Lukes (2005) says that command-obedience relationships — which would appear to fall into the category of an authoritative leadership coaching style — are “indispensable” to sports coaching. That may be true, but this analysis does confirm whether such a style has a measurable impact on Japanese player performance outcomes at the professional level.

Appendix

Tables and Figures

This appendix contains the figures and tables referenced above. Details and contexts to understanding the tables and figures are provided, along with sources and links to the Google Sheets files in which the numbers were prepared.



Findings	US born MLB	Findings	NPB	NPB - US born MLB	NPB/US born MLB
HBP/9	0.3936491442	HBP/9	0.6959766738	30.23%	1.768012668
BK/9	0.02534030317	BK/9	0.03909981313	1.38%	1.542989161
WP/9	0.3606807028	WP/9	0.5337124493	17.30%	1.479736634

[Figure 1/](#)[Table 1](#). Metrics of Control comparing U.S.-born MLB players to NPB pitchers from (2015–2022). U.S. data is from Stathead by Sports Reference, while NPB data is from Baseball Reference by Sports Reference. The preferred data would show metric/pitches thrown, however, pitches thrown could not be found and has been substituted with 9 innings of baseball played.

For example, HBP/9 is the number of batters hit for every 9 innings completed. A lower score is better.

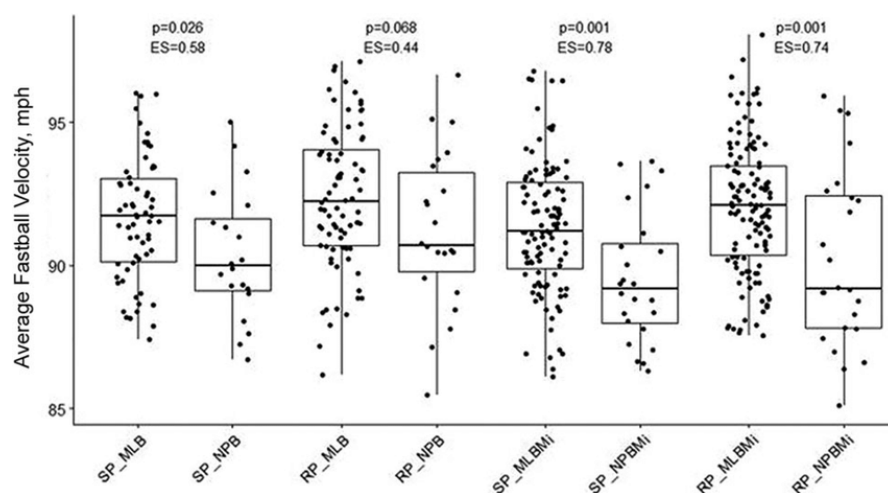


Figure 2. This figure (borrowed from Crotin et al. (2021)) shows boxplots of fastball velocity of starting and relieving pitchers who played in the Los Angeles Dodgers and the Hiroshima Toyo Carp organizations between 2015 and 2019. SP and RP notate starting vs relief pitchers, and “i” denotes minor league affiliate teams. A higher score is better.

NPB	values	Dif	%	US	values	Dif	%	Advantage
DP/IP	0.1833848073	-0.007853885215	-0.7854%	DP/IP*	0.1912386925	0.007853885215	0.7854%	US
PB/IP	0.01117853632	0.006130392625	0.6130%	PB/IP	0.005048143694	-0.006130392625	-0.6130%	US
Fld%	0.9861586595	0.0008235784341	0.0824%	Fld%	0.9853350811	-0.0008235784341	-0.0824%	NPB
*Data in the "US DP/IP" series is from all MLB players not just US born.								

[Table 2](#): Comparison of defensive performance between NPB players and U.S.-born MLB

players. From 2015-2022. NPB data from Baseball Reference by Sports Reference. U.S. data

from The Lahman Baseball Database. MLB data from Baseball Reference by Sports Reference.

MLB player birth country from Stathead by Sports Reference. Table finds the fld% of each group and adds up the PB and DP's divide them by the numbers of innings played to find the values. DP/IP = higher is better. PB/IP = lower is better. Fld% = higher is better. Advantage series notes which sample received the better score.

Year	Team USA	Team Japan	H2H (victor)	H2H (score)
2006	8	1	US	3-4
2009	4	1	JP	9-4
2013	6	3	N/A	N/A
2017	1	3	US	1-2
2023	2	1	JP	3-2
total (lower is better)	21	9	2-2	14-12
all data collected from Wikipedia				

[Table 3](#): World Baseball Classic performance for Team USA and Team Japan. Series two and three show the team's placement in that tournament. Data from Wikipedia.

values	ERA	W/L	WHiP	SO/BB	HBP	BK	WP	cFIP	Inn/GS*
US	4.215644092	0.4990072799	0.9568496823	2.650840925	0.2770615469	0.01783523144	0.2538574132	3.155076536	5.622946127
JP	3.989519522	0.5408921933	1.297994036	3.501126972	0.2375873287	0.01583915525	0.2807850248	3.136615248	5.741484185
diff(JP-US)	-0.2261245701	0.04188491336	0.3411443539	0.8502860472	-0.03947421825	-0.0019960762	0.02692761159	-0.01846128799	0.1185380579
%	-22.61%	4.19%	34.11%	85.03%	-3.95%	-0.20%	2.69%	-1.85%	11.85%
JP/US	0.9463606118	1.083936478	1.35652868	1.320760872	0.8575254535	0.8880824056	1.106073765	0.9941487036	1.02108113
Advantage	JP	JP	US	JP	JP	JP	US	JP	JP

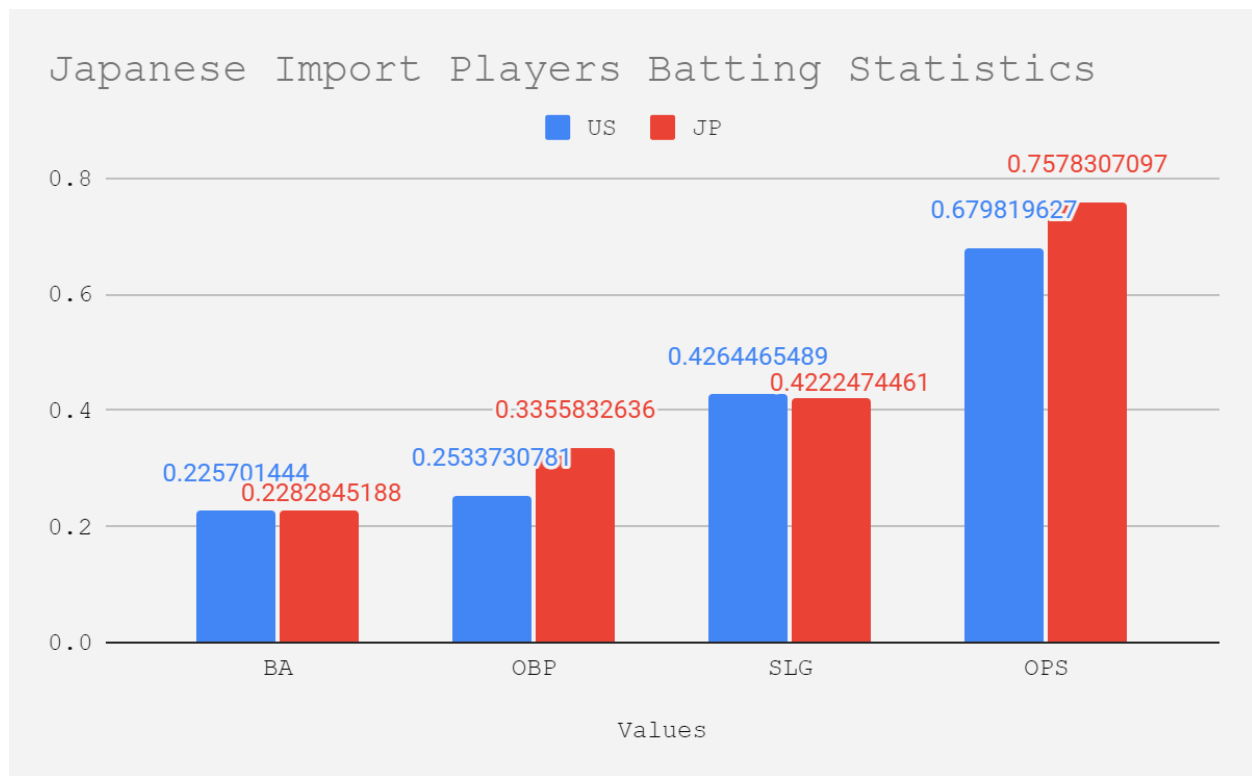
*series only includes data from pitchers in years where they started 100% of the games they played in that year.

[Table 4](#): Comparison of MLB pitching metrics between U.S. and Japanese born players, who played between 2015-2022. All numbers are the stat line of the samples as if each sample was a

singular player. data from Stathead by Sports Reference. Lower is better for ERA, WHiP, HBP, BK, WP, cFIP. Higher is better for W/L, K/BB, Inn/GS. Advantages row notes which sample had the better number.

Statcast metric	combined value*	MLB percentile	n**
mean_fielding_run_value	-1.0833	40.66	12/6
OAA	-15	40	12/3
arm_value	1	51.83	10/6
mean_arm_strength	81.78	37.2	5/5
JP Fld%	US Fld%	Difference (JP-US)	
0.9832953683	0.9853350811	-0.204%	
*OAA and arm value are sums of dataset while the rest are means.			
**(# of contributors to combined value)/(# of which qualified for %)			

[Table 5](#): Defensive performance of Japanese born MLB players, playing within the years of 2015-2022. Birthplace data from Stathead by Sports Reference, Fld% data from The Lahman Baseball Database, all other data from Baseball Savant. Only qualified fielders receive a percentile, so the “n” series shows the sample size of players that receive a value / how many of those qualified for a percentile.



Values	BA	OBP	SLG	OPS
US	0.225701444	0.2533730781	0.4264465489	0.679819627
JP	0.2282845188	0.3355832636	0.4222474461	0.7578307097
diff(JP-US)	0.002583074877	0.08221018551	-0.00419910279	0.07801108271
diff(rounded)	0.26%	8.22%	-0.0042	0.078
JP/US	1.011444654	1.324462986	0.9901532729	1.114752619
*two way players are in dataset, however only example is Shohei Ohtani				

[Figure 3/Table 6](#): comparison of batting performance between U.S. and Japanese born players in the MLB between 2015 and 2022, non-pitchers*, minimum of 100 plate appearances, every player has a different data point for every year qualified. Data from Stathead by Sports Reference.

player_name	ERA (NPB)	ERA (MLB)	ERA (Diff)	WHiP (NPB)	WHiP (MLB)	WHiP (diff)
Yu Darvish	1.44	3.9	2.46	0.828	1.28	0.452
Shohei Ohtani	3.2	3.31	0.11	1.263	1.161	-0.102
Masahiro Tanaka	1.27	2.77	1.5	0.943	1.056	0.113
Kenta Maeda	2.09	3.48	1.39	1.013	1.139	0.126
Yusei Kikuchi	3.08	5.46	2.38	1.033	1.515	0.482
Hisashi Iwakuma	2.42	3.16	0.74	1.05	1.277	0.227
Koji Uehara	3.81	4.05	0.24	1.182	1.245	0.063
Yoshihisa Hirano	2.67	2.44	-0.23	1.273	1.085	-0.188
Hirokazu Sawamura	6.08	3.06	-3.02	1.65	1.453	-0.197
Kazuhisa Makita	2.3	5.4	3.1	1.021	1.257	0.236
Shun Yamaguchi	2.91	8.06	5.15	1.159	1.753	0.594
Kohei Arihara	3.46	6.64	3.18	1.168	1.426	0.258
Kyuji Fujikawa	1.32	5.25	3.93	1.028	1.083	0.055
average	2.773076923	4.383076923	1.61	1.123923077	1.286923077	0.163

[Table 7](#): Comparison of MLB debut season with previous NPB season for Japanese pitchers

active in the MLB between 2015 and 2022. Sample from Stathead by Sports Reference, MLB

data from Baseball Reference by Sports Reference, and NPB data from ProEyeKyu

player_name	league+year	BA	OBP	SLG	OPS
Nori Aoki	NPB(2011)	0.292	0.358	0.36	0.718
	MLB(2012)	0.285	0.349	0.36	0.71
	diff	-0.007	-0.009	0	-0.008
Ichiro Suzuki	NPB(2000)	0.387	0.46	0.539	0.999
	MLB(2001)	0.35	0.381	0.457	0.838
	diff	-0.037	-0.079	-0.082	-0.161
Seiya Suzuki	NPB(2021)	0.317	0.433	0.639	1.072
	MLB(2022)	0.262	0.336	0.433	0.77
	diff	-0.055	-0.097	-0.206	-0.302
Yoshi Tsutsugo	NPB(2019)	0.272	0.388	0.511	0.899
	MLB(2020)	0.197	0.314	0.395	0.708
	diff	-0.075	-0.074	-0.116	-0.191
Shogo Akiyama	NPB(2019)	0.303	0.392	0.471	0.863
	MLB(2020)	0.245	0.357	0.297	0.654
	diff	-0.058	-0.035	-0.174	-0.209
Munenori Kawasaki	NPB(2011)	0.267	0.31	0.327	0.637
	MLB(2012)	0.192	0.257	0.202	0.459
	diff	-0.075	-0.053	-0.125	-0.178
Shohei Ohtani	NPB(2017)	0.332	0.403	0.54	0.943
	MLB(2018)	0.285	0.361	0.564	0.925
	diff	-0.047	-0.042	0.024	-0.018

[Table 8](#): Comparison of MLB debut season with previous NPB season for Japanese hitters active in the MLB between 2015 and 2022. Sample from Stathead by Sports Reference, MLB data from Baseball Reference by Sports Reference, and NPB data from ProEyeKyuu.

Endnotes

1. (ERA) Earned Run Average: $((9 * \text{Earned Runs}) / \text{Innings Pitched})$. ERA is a pitching statistic that shows the average number of earned runs² that a pitcher gives up every nine innings of work they complete. Lower is better.
2. (ER) Earned Runs: ERs are runs scored without the benefit of an error³ or passed ball⁴. The difference between earned and unearned runs has no effect on the score of the game; the distinction exists for statistics.
3. (E) Error: An error refers to when an official judges that a fielder fails to make a play that they should have made. Common examples include dropping or failing to catch a ball a fielder is near to, and throwing a ball that does not reach its intended target.
4. (PB) Passed Ball: A passed ball is when a runner advances because a catcher was unable to catch/hold on to a pitch that an official judges that they should have.
5. (WHiP) Walks plus Hits per Inning Pitched: $((\text{Walks} + \text{Hits}) / \text{Innings Pitched})$. WHiP is a pitching statistic that shows the expected number of baserunners a pitcher allows every inning. Lower is better.
6. (K/BB ratio) Strikeout-to-walk ratio: $(\text{Strikeouts} / \text{Walks})$. K/BB, often shown as SO/BB ratio, is a pitching statistic that measures how many strikeouts a pitcher records per walk given out. It is often displayed in fraction form but, when simplified, higher is better.
7. (HBP) Hit by Pitch. A hit by pitch is when the pitcher hits the batter with the ball, in which case the batter is awarded first base. In order to count as a HBP, the pitch must have been out of the strike zone, the batter must have been in the batter's box, the batter must not have swung at the pitch, and the batter must not have intentionally let the ball hit him (determined by judgement of officials).

8. (WP) Wild Pitch: A wild pitch is when a runner advances because a catcher was unable to catch/hold on to a pitch that an official judges that they were not expected to have been able to stop.
9. (BK) Balk: A balk is an illegal movement made by a pitcher on the mound with the intention of deceiving baserunners. When a balk occurs, all base runners advance a base.
10. (OAA) Outs Above Average: OAA Is a StatCast era statistic. For every ball hit in play, StatCast data determines probability that the average player makes the out. If the fielder makes the play, then they are awarded the chance the average player does not make the out. If they do not make the out, then they lose the chance the average player makes it. For example, if the catch probability of a ball in play is 90% and the fielder records the out, then they receive .1 OAA. If the fielder fails, then they lose .90 OAA. Higher is better.
11. (Fld%) Fielding Percentage: $((\text{Putouts} + \text{Assists}) / (\text{Putouts} + \text{Assists} + \text{Errors}))$ Fld% is a defensive statistic that tracks how often a player touches a ball in play and does not commit an error. Putouts are when a fielder makes an out. Assists are when a fielder touches a ball before a different player makes an out.
12. (DP) Double Play: A double play is when two outs are collected on a single plate appearance.
13. (cFIP) Fielding Independent Pitching Constant: $(\text{lgERA} - (((13 * \text{lgHR}) + (3 * (\text{lgBB} + \text{lgHBP})) - (2 * \text{lgK})) / \text{lgIP}))$ cFIP is the league constant used for calculating FIP. FIP is a statistic used to determine a pitcher's performance in situations where the defense has no control over the outcome of a plate appearance. FIP formula = $((13 * \text{HR}) + (3 * (\text{BB} + \text{HBP})) - (2 * \text{K})) / \text{IP} + \text{cFIP}$

14. (BA) Batting Average: (H/AB) BA is how often a player is able to get a hit. In 2024 the MLB league average was $\sim .243$. BA is hits per AB (at bats) not PA (plate appearances). PA is the number of times the hitter came to the plate, while AB is the number of PAs that did not result in a BB, HBP, sacrifice out (when a ball is hit in play for an out but a run scores), catcher's interference, and dropped third strike. This is done so that BA does not go down when a batter reaches base without putting a ball in play, or intentionally makes an out.
15. (OBP) On Base Percentage: $((\text{Hits} + \text{Walks} + \text{Hit by Pitch}) / (\text{At Bats} + \text{Walks} + \text{Hit by Pitch} + \text{Sacrifice Flies}))$ OBP is how often a hitter reaches base. Reaching base by error or fielder's choice (when an out is collected but on a different base runner) counts against OBP, while reaching base by catcher interference, or dropped third strike has no effect on OBP.
16. (SLG) Slugging: $((1B + 2Bx2 + 3Bx3 + HRx4) / AB)$ SLG is a variation of BA that is weighted for which base the hitter reaches on each hit.
17. (OPS) On base plus slugging: $((OBP + SLG)/2)$ OPS averages out the hitters OBP and SLG. The purpose is to have a hitter statistic that values the ability to reach base and hit for power.
18. Fielding run value: fielding value is a StatCast metric made by Baseball Savant, which evaluates a player's defensive impact.
19. Arm value: arm value is a StatCast metric made by Baseball Savant that evaluates how well a defender can prevent a baserunner from advancing.

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